GENERIC FLATNESS STRATIFICATION FOR A FINITELY PRESENTED MODULE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 15579-15612, immediately following the statement at 15565-15577.

Editorial note. Four cross-chapter pointers (Limits, Morphisms) are recorded in assembly.external_references.

Editorial note. The statement's descending chain with its three conditions is retained verbatim in the statement span.

Allowed standard prerequisites (editorial).

- Generic flatness over a reduced base. Morphisms, proposition morphisms-proposition-generic-flatness-reduced, cited at proof lines 15602-15604. Supplies the dense open U_0 over which the pullback of F is flat, which drives the descending chain.
- Approximation of a finitely presented morphism. Limits, proposition limits-proposition-approximate with the descent lemmas cited in the proof. Reduces the general statement to a Noetherian situation where the chain terminates.
- Descending finite presentation and finite modules. Limits, lemmas limits-lemma-descend-finite-presentation and limits-lemma-descend-modules-finite-presentation, cited in the proof. Keeps the finiteness hypotheses along the approximation.
- Descending chains of closed subsets in a Noetherian space. used at proof lines 15609-15611. Gives termination of the constructed chain, producing the required t.

Source reference. Original reference 01ZM: focused-seeds/geometry/S9-morph-generic-flatness-stratification/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZR: focused-seeds/geometry/S9-morph-generic-flatness-stratification/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZA: focused-seeds/geometry/S9-morph-generic-flatness-stratification/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 052B: focused-seeds/geometry/S9-morph-generic-flatness-stratification/ref-tags.txt, source lines 8–8. Navigation pointer only: the

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counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Generic flatness stratification)**.** *Let $f : X \to S$ be a morphism of finite presentation between quasi-compact and quasi-separated schemes. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module of finite presentation. Then there exists a $t \geq 0$ and closed subschemes*

$$S \supset S_0 \supset S_1 \supset \ldots \supset S_t = \emptyset$$

such that $S_i \to S$ is defined by a finite type ideal sheaf, $S_0 \subset S$ is a thickening, and $\mathcal{F}$ pulled back to $X \times_S (S_i \setminus S_{i+1})$ is flat over $S_i \setminus S_{i+1}$.

Proof. We can find a cartesian diagram

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow & & \downarrow \\ S & \longrightarrow & S_0 \end{array}$$

and a finitely presented $\mathcal{O}_{X_0}$-module $\mathcal{F}_0$ which pulls back to $\mathcal{F}$ such that X_0 and S_0 are of finite type over $\mathbf{Z}$. See Limits, Proposition 01ZA and Lemmas 01ZM and 01ZR. Thus we may assume X and S are of finite type over $\mathbf{Z}$ and $\mathcal{F}$ is a coherent $\mathcal{O}_X$-module.

Assume X and S are of finite type over $\mathbf{Z}$ and $\mathcal{F}$ is a coherent $\mathcal{O}_X$-module. In this case every quasi-coherent ideal is of finite type, hence we do not have to check the condition that S_i is cut out by a finite type ideal. Set $S_0 = S_{red}$ equal to the reduction of S. By generic flatness as stated in Morphisms, Proposition 052B there is a dense open $U_0 \subset S_0$ such that $\mathcal{F}$ pulled back to $X \times_S U_0$ is flat over U_0. Let $S_1 \subset S_0$ be the reduced closed subscheme whose underlying closed subset is $S \setminus U_0$. We continue in this way, provided $S_1 \neq \emptyset$, to find $S_0 \supset S_1 \supset \ldots$. Because S is Noetherian any descending chain of closed subsets stabilizes hence we see that $S_t = \emptyset$ for some $t \geq 0$. $\square$